

HW9

Group 12 - Shir, Omer & Ido

16/06/26

Group:

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Paths and Data

```
if (!require('data.table')) { install.packages('data.table'); library('data.table') }
if (!require('splines'))      { install.packages('splines');   library('splines') }
if (!require('MASS'))         { install.packages('MASS');      library('MASS') }
```

```
source("../utils.R")
source("../lab5/lab5_utils.R")
```

```
healthy_file <- "../lab2/data/TCGA-13-0723-10B_lib2_all_chr1.forward"
chr1_file    <- "../lab1/data/chr1_str.rda"
```

```
healthy_reads <- fread(healthy_file, col.names = c("Chr", "Loc", "Length"))
load(chr1_file)
```

```
BIN <- 2500
```

```
# Two arms around centromere (120-130 Mb excluded)
dfA <- bin_data(healthy_reads, chr1, 5000001L, 119990000L, BIN)
dfB <- bin_data(healthy_reads, chr1, 130000001L, 242000000L, BIN)
```

```
add_pos <- function(df, beg) {
  df$pos <- beg + (seq_len(nrow(df)) - 1) * BIN + BIN / 2
  df
}
dfA <- add_pos(dfA, 5000001L)
dfB <- add_pos(dfB, 130000001L)
df <- rbind(dfA, dfB)
```

```
# GC spline + plug-in estimator (same as lab 8 method b)
eps <- 1
nz <- df$reads > 0
fit <- fit_gc_spline(df[nz, ])
```

```
fhat <- pmax(predict(fit, data.frame(gc = df$gc)), 0)

## Warning in bs(gc, degree = 3L, knots = c(`25%` = 0.3684, `50%` = 0.4064, : some
## 'x' values beyond boundary knots may cause ill-conditioned bases

raw <- (df$reads + eps) / (fhat + eps)
df$cn <- raw / median(raw[nz])

cat(sprintf("Total bins: %d | non-zero: %d\n", nrow(df), sum(nz)))

## Total bins: 90796 | non-zero: 85048
```

Part A – Test Statistic

We need a statistic T that summarises whether a genomic region has more or fewer reads than expected under copy number 1, while being **resistant to single noisy bins** – meaning one extreme bin should not be able to drive the entire region to significance on its own.

We work on $\log(\hat{a}_k)$, the log of the GC-corrected copy-number estimate from Lab 8 (method b, single-sample spline correction). Taking the log has two advantages: under H_0 (CN = 1) the statistic is centred at 0, and the distribution is approximately symmetric (raw $\hat{a}_k$ is right-skewed since CN cannot go below 0). Gains produce positive values; losses produce negative values.

We aggregate over a sliding window of $W = 11$ bins (≈ 27.5 kb) and compare three candidates:

Statistic	Formula	Robustness
S1: plain log	$\log(\hat{a}_k)$ – per single bin, no aggregation	None – baseline
S2: trimmed mean	$\log(\hat{a})_{10\%}$ over window	Hard-discards top/bottom 10% of bins
S3: Huber location	$\hat{\mu}_{\text{Huber}}(\log(\hat{a}), k = 1.345)$ over window	Continuously downweights outlier bins

S1 serves as a baseline to show what robustness buys us. Both S2 and S3 satisfy the requirement: no single bin can dominate the window-level statistic. S3 (Huber) is the preferred choice because at $k = 1.345$ it achieves 95% efficiency under normality – no power is lost when the data are well-behaved – while degrading gracefully otherwise. S2's hard trim boundary discards information for bins near the cutoff.

```
W <- 11
HALF <- (W - 1) / 2

log_cn <- log(pmax(df$cn, 1e-6)) # log(a-hat), floored to avoid -Inf

# Rolling window helper: applies FUN to each bin's W-neighbourhood
roll_stat <- function(x, fun) {
  out <- rep(NA_real_, length(x))
  for (i in seq_along(x)) {
    idx <- max(1, i - HALF) : min(length(x), i + HALF)
    sub <- x[idx][df$reads[idx] > 0] # only non-zero bins in window
    if (length(sub) >= 5) out[i] <- fun(sub)
  }
  out
}
```

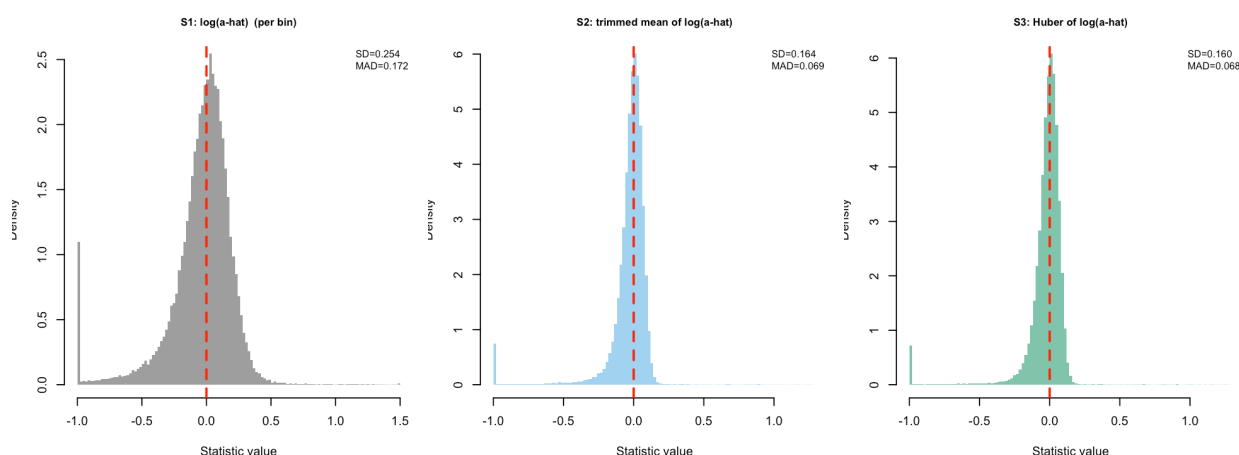
```

df$S1 <- log_cn # plain log(a-hat)
df$S2 <- roll_stat(log_cn, function(x) mean(x, trim = 0.10)) # trimmed mean
df$S3 <- roll_stat(log_cn, function(x) huber(x, k = 1.345)$mu) # Huber

par(mfrow = c(1, 3), mar = c(4, 3.5, 2.5, 1))
stats <- list(
  list(v = df$S1[nz], lab = "S1: log(a-hat) (per bin)",
    list(v = df$S2[nz & !is.na(df$S2)], lab = "S2: trimmed mean of log(a-hat)",
      list(v = df$S3[nz & !is.na(df$S3)], lab = "S3: Huber of log(a-hat)"))
)
cols <- c("#555555", "#56B4E9", "#009E73")

for (i in seq_along(stats)) {
  v <- stats[[i]]$v
  v <- pmin(pmax(v, -1), 1.5)
  hist(v, breaks = 100, freq = FALSE, col = adjustcolor(cols[i], 0.6), border = NA,
    main = stats[[i]]$lab, xlab = "Statistic value", cex.main = 0.85)
  abline(v = 0, col = "red", lwd = 2, lty = 2)
  legend("topright", bty = "n", cex = 0.8,
    legend = sprintf("SD=%.3f\nMAD=%.3f", sd(v, na.rm=TRUE), mad(v, na.rm=TRUE)))
}

```



S2 and S3 are visibly tighter than S1 (lower SD/MAD) while remaining centred at 0 – robustness without bias.

Part B – Negative Control Regions and Null Distribution

How we select the negative controls. We need genomic regions where $CN = 1$ with high confidence, so the statistic values there serve as our empirical H_0 distribution. We use the paired healthy tissue sample (TCGA-13-0723-10B): a normal blood cell should have $CN = 1$ almost everywhere on chr1, making the healthy genome our natural candidate pool. We compute the GC-corrected estimator on the healthy sample and filter for bins that are genuinely undisturbed, comparing two strategies:

- **Strategy A – global MAD filter:** Keep every non-zero bin whose GC-corrected CN is within 3 MAD of 1. A loose global threshold; ~91% of bins pass. It removes only extreme outliers; everything else is treated as a control.
- **Strategy B – local stability filter:** Keep a bin only if its 11-bin neighbourhood simultaneously has

a median near 0 in log-space ($|\tilde{T}| \leq 0.095$) **and** a low local SD (≤ 0.14). Stricter (~28% of bins pass): a bin is accepted only if the whole local region is quiet. Bins at a CN-event boundary – even if their own value is near 1 – are excluded because their neighbours reveal the disruption.

How we model H_0 . We do not assume a parametric form for H_0 . The statistic values from the selected negative control bins *are* the empirical null distribution – we treat them as a direct sample from the distribution of T under the null hypothesis. The histograms and QQ plots below characterise this distribution for each combination of statistic and strategy. The key diagnostic is normality: a linear QQ plot means that both empirical and parametric (z-score) p-values will agree, which is a useful consistency check.

How p-values will be derived (Part C). For any test bin k , the two-sided empirical p-value is the fraction of null statistic values at least as extreme:

$$p_k = \frac{|\{j \in \text{ctrl} : |T_j| \geq |T_k|\}|}{|\text{ctrl}|}$$

Because the null is approximately normal (confirmed below), this is equivalent to $p_k = 2\Phi(-|z_k|)$ where $z_k = T_k/\hat{\sigma}_0$ and $\hat{\sigma}_0$ is the SD of the control bins. Both approaches give consistent results; the empirical version makes no distributional assumption.

```
# Strategy A: global MAD-3
outlier_thresh <- 3 * mad(df$cn[nz], constant = 1)
stratA <- nz & abs(df$cn - 1) <= outlier_thresh

# Strategy B: local stability in log space
win_med_log <- roll_stat(log_cn, median)
win_sd_log <- roll_stat(log_cn, sd)
stratB <- nz & !is.na(win_med_log) & !is.na(win_sd_log) &
  abs(win_med_log) <= 0.095 & win_sd_log <= 0.14

cat(sprintf("Strategy A: %d bins (%.0f%% of non-zero)\n",
  sum(stratA), 100 * sum(stratA) / sum(nz)))

## Strategy A: 77185 bins (91% of non-zero)

cat(sprintf("Strategy B: %d bins (%.0f%% of non-zero)\n",
  sum(stratB), 100 * sum(stratB) / sum(nz)))

## Strategy B: 23639 bins (28% of non-zero)

stat_list <- list(S1 = df$S1, S2 = df$S2, S3 = df$S3)
stat_labs <- c("S1: log(a-hat)", "S2: trimmed mean", "S3: Huber")
strat_list <- list(A = stratA, B = stratB)
strat_labs <- c("Strategy A (MAD-3)", "Strategy B (local stability)")
strat_cols <- c("#56B4E9", "#009E73")

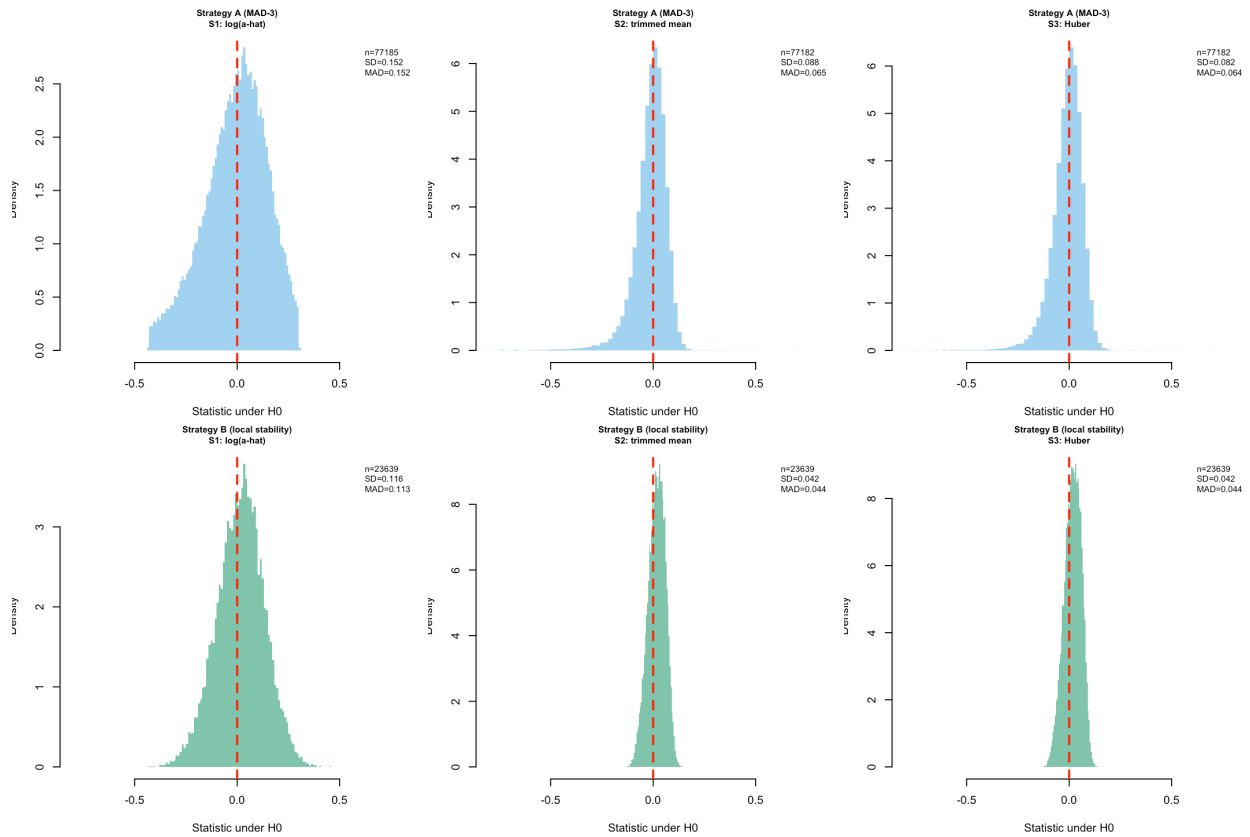
par(mfrow = c(2, 3), mar = c(4, 3.5, 2.5, 1))

for (si in 1:2) {
  for (ti in 1:3) {
    mask <- strat_list[[si]] & !is.na(stat_list[[ti]])
    v <- stat_list[[ti]][mask]
    v <- pmin(pmax(v, -1), 1.5)
    hist(v, breaks = 80, freq = FALSE,
      col = adjustcolor(strat_cols[si], 0.6), border = NA,
      main = sprintf("%s\n%s", strat_labs[si], stat_labs[ti]),
      xlab = "Statistic under H0", cex.main = 0.78,
      xlim = c(-0.8, 0.8))
  }
}
```

```

abline(v = 0, col = "red", lwd = 2, lty = 2)
legend("topright", bty = "n", cex = 0.75,
      legend = sprintf("n=%d\nSD=%.3f\nMAD=%.3f",
                        length(v), sd(v, na.rm=TRUE), mad(v, na.rm=TRUE)))
}
}

```

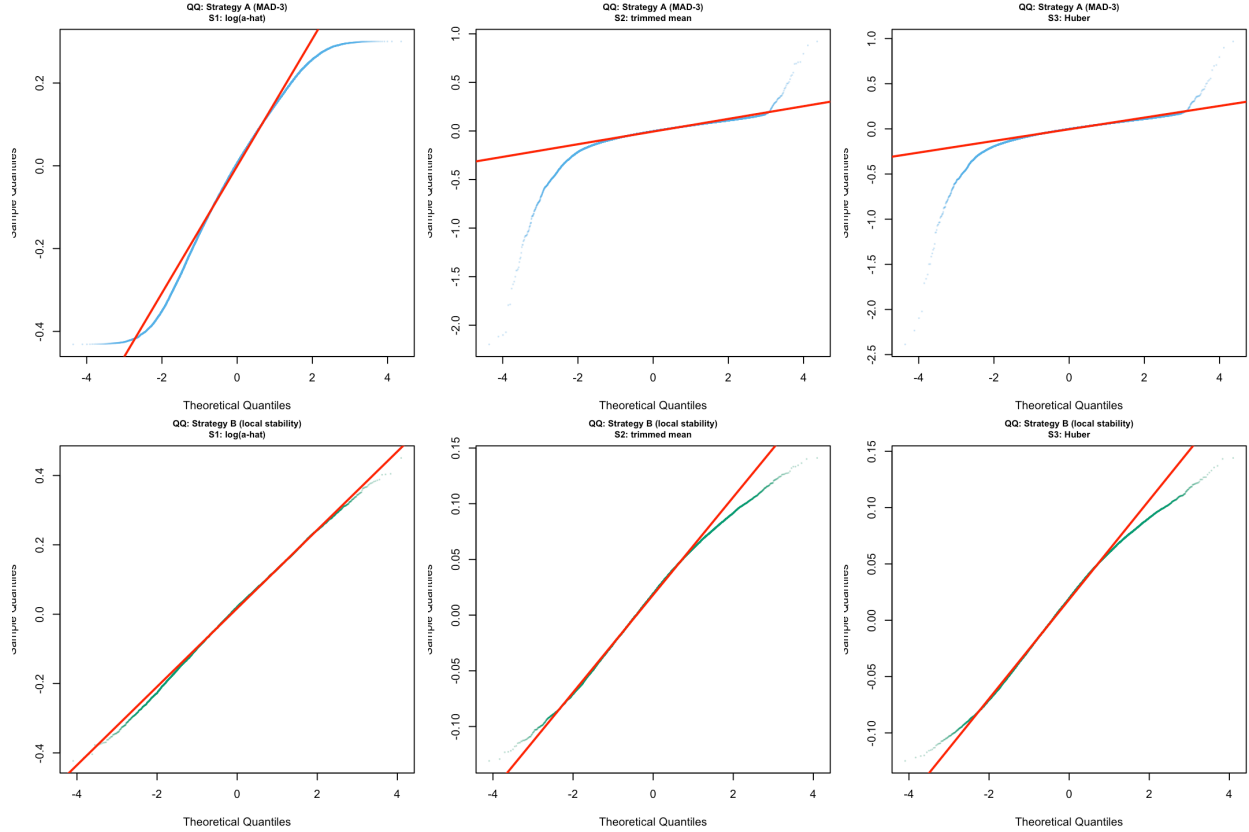


```

par(mfrow = c(2, 3), mar = c(4, 3.5, 2.5, 1))

for (si in 1:2) {
  for (ti in 1:3) {
    mask <- strat_list[[si]] & !is.na(stat_list[[ti]])
    v <- stat_list[[ti]][mask]
    qqnorm(v, pch = 20, cex = 0.2, col = adjustcolor(strat_cols[si], 0.3),
          main = sprintf("QQ: %s\n%s", strat_labs[si], stat_labs[ti]),
          cex.main = 0.78)
    qqline(v, col = "red", lwd = 2)
  }
}

```



```
rows <- list()
for (si in 1:2) {
  for (ti in 1:3) {
    mask <- strat_list[[si]] & !is.na(stat_list[[ti]])
    v <- stat_list[[ti]][mask]
    rows[[length(rows)+1]] <- data.frame(
      Strategy = names(strat_list)[si],
      Statistic = names(stat_list)[ti],
      n = length(v),
      Mean = round(mean(v), 4),
      SD = round(sd(v), 4),
      MAD = round(mad(v), 4)
    )
  }
}
summary_tbl <- do.call(rbind, rows)
knitr::kable(summary_tbl, caption = "Null distribution summary: 3 statistics x 2
  ↪ strategies")
```

Table 1: Null distribution summary: 3 statistics x 2 strategies

Strategy	Statistic	n	Mean	SD	MAD
A	S1	77185	-0.0074	0.1518	0.1525
A	S2	77182	-0.0145	0.0907	0.0646
A	S3	77182	-0.0108	0.0841	0.0640
B	S1	23639	0.0163	0.1155	0.1133
B	S2	23639	0.0171	0.0417	0.0437

Strategy	Statistic	n	Mean	SD	MAD
B	S3	23639	0.0172	0.0416	0.0440

Conclusion. Strategy B + S3 (Huber) gives the tightest, most normal null ($SD = 0.042$, nearly linear QQ plot) and is our chosen combination for parts C onward. Strategy A + S1 serves as a baseline that makes the benefit of both robustness and stricter control selection clearly visible.

External sources statement

We used AI (Claude) to design the negative control selection strategies (exploratory analysis comparing MAD-3 vs local stability filters) and to structure this document. Statistical choices (Huber $k=1.345$, trimming fraction, window width) and interpretation are our own.